

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2023-2024
M.Tech. in AIML
End-Semester Test
(EC-2 Regular Paper)

Course No. : AIMLZG525
Course Title : Natural Language Processing
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : October-2024

No. of Pages	= 4
No. of Questions	= 7

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Question 1. [5 Marks]

You have the following observed words and corresponding emission probabilities for an HMM POS tagger:

Word	Noun	Verb	Adj
"cat"	0.9	-	0.1
"jumps"	0.3	0.7	-
"quickly"	-	0.3	0.7

Question:

Using the forward algorithm, calculate the probability of the observation sequence "cat jumps quickly" given the following transition probabilities:

Transition	Noun → Noun	Noun → Verb	Verb → Verb	Verb → Adj
Probability	0.5	0.5	0.7	0.3

Solution:

1. Initialization:

Initial state is equally likely for Noun or Verb, so ($P(\text{text}\{\text{Noun}\}) = 1/2$), ($P(\text{text}\{\text{Verb}\}) = 1/2$).

2. Forward Step 1 ("cat"):

($P(\text{text}\{\text{Noun}\}) = 1/2$ times $0.9 = 0.45$)

Verb is not an option since the emission probability for "cat" as a verb is 0.

3. Forward Step 2 ("jumps"):

($P(\text{text}\{\text{Verb}\}) = P(\text{text}\{\text{Noun}\})$ times $P(\text{text}\{\text{Noun} \rightarrow \text{Verb}\})$ times $P(\text{text}\{\text{Verb emits "jumps"}\})$)

($P(\text{text}\{\text{Verb}\}) = 0.45$ times 0.5 times $0.7 = 0.1575$)

4. Forward Step 3 ("quickly"):

($P(\text{text}\{\text{Adj}\}) = P(\text{text}\{\text{Verb}\})$ times $P(\text{text}\{\text{Verb} \rightarrow \text{Adj}\})$ times $P(\text{text}\{\text{Adj emits "quickly"}\})$)

($P(\text{text}\{\text{Adj}\}) = 0.1575$ times 0.3 times $0.7 = 0.0331$)

5. Conclusion:

The final probability of the sequence "cat jumps quickly" is 0.0331.

Question 2. [5 Marks]

For an HMM with two hidden states S1 and S2, and the following transition and emission probabilities:

Transition Probabilities			
	S1	S2	
S1	0.7	0.3	
S2	0.6	0.4	

Emission Probabilities			
	S1	S2	
"X"	0.6	0.4	
"Y"	0.3	0.7	

Question:

Use the Forward Algorithm to calculate the probability of the observation sequence "X Y" given the initial state probabilities ($P(S1) = 0.6$) and ($P(S2) = 0.4$).

Solution:

1. Initialization:

- For (S1): ($P(S1, X) = 0.6 \text{ times } 0.6 = 0.36$)
- For (S2): ($P(S2, X) = 0.4 \text{ times } 0.4 = 0.16$)

2. Forward Step (Observation "Y"):

- For (S1) (transitioning from S1 and S2):
($P(S1) = (0.36 \text{ times } 0.7 \text{ times } 0.3) + (0.16 \text{ times } 0.4 \text{ times } 0.3) = 0.0756$)
- For (S2) (transitioning from S1 and S2):
($P(S2) = (0.36 \text{ times } 0.3 \text{ times } 0.7) + (0.16 \text{ times } 0.6 \text{ times } 0.7) = 0.1032$)

3. Conclusion:

The total probability of the observation sequence "X Y" is

Question 3. [6 Marks]

Consider the statement: "The extremely hot desert is a hostile place for living"

- Using **Depth First, Simple Top Down Parsing** method, work out the steps **only till** the phrase "*The extremely hot desert*" is successfully parsed. Provide one line explanation for each step.

The grammar rules to be applied and the lexicon are given below.

CFG Rules				Lexicon			
Rule No	Grammar	Rule No	Grammar	Word	Category	Word	Category
1	$S \rightarrow NP VP$	10	$DET \rightarrow 'a'$	The	DET	a	DET
2	$NP \rightarrow DET AP N$	11	$ADV \rightarrow 'extremely'$	extremely	ADV	hostile	ADJ
3	$NP \rightarrow DET N$	12	$ADJ \rightarrow 'hot'$	hot	ADJ	place	N
4	$AP \rightarrow ADV ADJ$	13	$ADJ \rightarrow 'hostile'$	desert	N	for	PREP
5	$AP \rightarrow ADJ$	14	$N \rightarrow 'desert'$	is	V	living	N ADJ
6	$VP \rightarrow V NP$	15	$N \rightarrow 'place'$				
7	$VP \rightarrow V NP PP$	16	$PREP \rightarrow 'for'$				
8	$PP \rightarrow PREP NP$	17	$V \rightarrow 'is'$				
9	$DET \rightarrow 'The'$	18	$N \rightarrow 'living'$				

SOLUTION

(Give 1 mark each if the steps at SNo. 2,3,4,5,6,7 (including Backup) are correctly identified)

1The 2extremely 3hot 4desert 5is 6a 7hostile 8place 9for 10living11

SNo.	State	Backup	Remark
1.	((S) 1)		Start of the process
2.	((NP VP) 1)		Rule 1
3.	((DET AP N VP) 1)	((DET N VP) 1)	NP expanded: Rule 2 and 3
4.	((AP N VP) 2)	((DET N VP) 1)	'The' matches DET: Rule 9
5.	((ADV ADJ N VP) 2)	((AP ADJ N VP) 2) ((DET N VP) 1)	AP expanded: Rule 4 and 5
6.	((ADJ N VP) 3)	((AP ADJ N VP) 2) ((DET N VP) 1)	'extremely' matches ADV: Rule 11
7.	((N VP) 4)	((AP ADJ N VP) 2) ((DET N VP) 1)	'hot' matches ADJ: Rule 12
8.	((VP) 5)	((AP ADJ N VP) 2) ((DET N VP) 1)	'desert' matches N: Rule 14
9.			'The extremely hot desert' matches NP

Question 4. [3 + 3 = 6 Marks]

- In the context of **Dependency Parsing** differentiate between the **Transition-based method** and the **Graph-based method** using the following aspects: i) The main ideas behind the approach ii) Name(s) of the algorithm(s) used [3 Marks]
- What will be the Constituency Parse Tree and the Dependency Parse Tree for the statement "**The desert became very hot**"? (Only show the final tree, derivation steps are not expected. Use appropriate grammar rules.) [3 Marks]

In the constituency Parse tree, relevant rules among the following may be used:

S → NP VP
 NP → Det N
 VP → V AdvP
 VP → V AdjP
 AdjP → Adv Adj
 VP → V NP
 VP → V PP
 AdvP → Adv

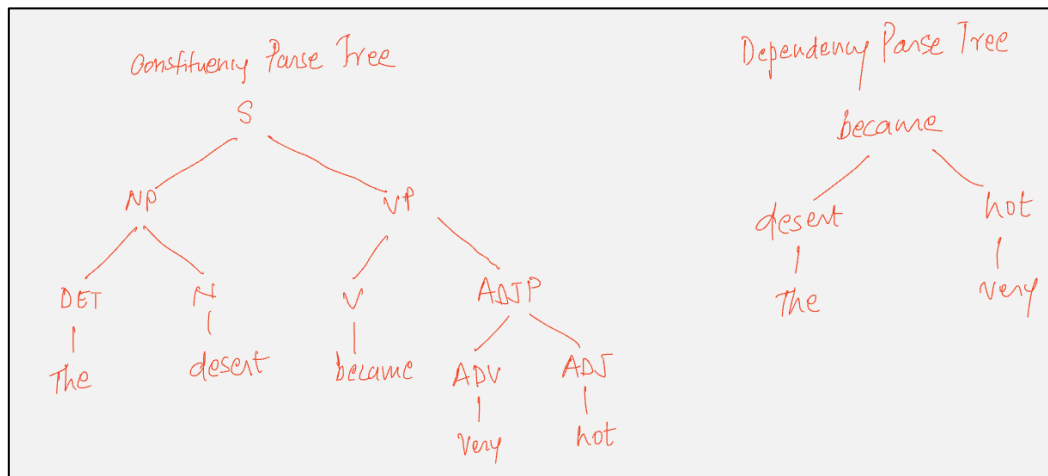
SOLUTION

4(a) - Give total 2 marks for correct 'main ideas' and 1 mark if algorithms are correct. See the Table below for possible answers to both these aspects. Note that the answer to 'the main idea' may include a combination of the aspects listed below.

Give partial marks (with appropriate justification) if the answers are partially correct.

Aspect	Transition-Based Parsing	Graph-Based Parsing
Parsing Approach	Builds the parse tree incrementally by performing transitions involving a stack, a buffer and Arcs states.	Treats the dependency parsing problem as finding the Maximum SPanning Tree (MST) in a complete graph.
Tree Construction	Parses the sentence word by word, adding dependencies between words as transitions are made.	Constructs a complete dependency graph and selects the optimal tree using graph algorithms.
Parsing Process	Works sequentially, making one decision at a time based on the current state (stack/buffer).	Scores all possible dependency arcs at once, then extracts the highest scoring tree.
Efficiency	Typically faster due to the incremental nature of parsing, especially useful for online parsing.	Slower in comparison due to the need to evaluate all possible edges before finding the optimal tree.
Algorithm Examples	Arc-Standard, Arc-Eager	MSTParser, Chu-Liu/Edmonds Algorithm
Use Cases	Suitable for real-time and low-latency parsing tasks. Often used in speech-to-text or interactive systems.	More suitable for tasks requiring global accuracy, such as document parsing, where speed is less critical.

4(b) Constituency Parse Tree and Dependency Parse Tree for the sentence "**The desert became very hot**". (1.5 marks per tree. Give partial marks if partially correct)



Question 5. [4+2=6 Marks]

Draw the knowledge graph that shows an example of a derived inference.

2 marks

RDF Statements:

ex:LivingThing rdfs:subClassOf ex:Object .

ex:Animal rdfs:subClassOf ex:LivingThing .

ex:Person rdfs:subClassOf ex:Animal .

ex:Faculty rdf:type ex:Person

ex:Person rdf:hasEmail ex:EmailId

ex:Course rdf:taughtBy "Chetana Gavankar"

ex:"Chetana Gavankar" rdf:EmailId ex:"chetana.gavankar@pilani.bits-pilani.ac.in"

Consider the following sentence: "I need to fix the bug in my code." The following entries are available in a dictionary for the word "bug":

4 marks

Word	Gloss	Usage Examples
bug (1)	A small insect	Insecticides can fix proliferation of insects
bug (2)	A flaw in a program that causes it to behave incorrectly	Fixing defects in code after testing is one of the steps in the Software Development Life Cycle.

We need to disambiguate the word "bug" in the above sentence, using the Lesk algorithm. Explain the steps to be followed.

Question 6. [2 + 4= = 6 Marks]

a) Consider following contextual embeddings for the word "bank" for each of its senses:

bank _N ¹	Financial institution	[1,4,1]
bank _N ²	Sloping land	[2,8,1]
bank _V ¹	Tip laterally	[4,7,3]
bank _V ³	Do business with a bank	[3,3,3]

The contextual embedding for the target word "bank" in a specific sentence was derived as [2,6,2]. What is the sense of the target word?

Solution:

We find the cosine similarity between target word and each sense.

Cosine Similarity = Dot Product of Vectors / Product of magnitude of each vector

Sense	Magnitude	Target	Target Magnitude	Dot Product	Product of Magnitudes	Cosine Similarity
1 4 1	4.24	2 6 2	6.633249581	28	28.14	0.995
2 8 1	8.31			54	55.10	0.980
4 7 3	8.60			56	57.06	0.981
3 3 3	5.20			30	34.47	0.870

Hence the Answer is sense "Financial Institution" or bank_N¹
(2 marks)

- b) Given the sentence: “**Stay Curious**” and information below, demonstrate the mathematical application of a single attention head to derive Self Attention. Support your answer with diagrams and relevant mathematical equations, detailing all steps. Apply score normalization, masking and softmax.

Embeddings:

	Embedding
Stay	[1 1]
Curious	[1 1]

Weight Matrix for Query W^q

[1 0
0 2]

Weight Matrix for Keys W^k

[0 2
1 0]

Weight Matrix for Values W^v

[2 3
3 2]

- $\text{Exp}(3.54) = 34.31$
- Compute all fractions only to Second Decimal place (X.XX)

Solution:

(Marking scheme in brackets against each step)

1. (0.5 marks)

	Input
Stay	[1 1]
Curious	[1 1]

2. Compute Q, K & V

(1 marks)

$$\mathbf{Q} = \mathbf{XW}^q; \mathbf{K} = \mathbf{XW}^k; \mathbf{V} = \mathbf{XW}^v$$

Q

1	2
1	2

K

1	2
1	2

V

5	5
5	5

3. Compute $\mathbf{QK}^T$

(0.5 marks)

5	5
5	5

4. Apply Normalization with $d_k=2$

(0.5 marks)

$$\frac{QK^T}{\sqrt{d_k}}$$

3.54	3.54
3.54	3.54

5. Apply Masking

(0.5 marks)

$$\text{mask} \left(\frac{QK^T}{\sqrt{d_k}} \right)$$

3.54	-1.00E+99
3.54	3.54

6. Apply Softmax

(0.5 marks)

$$\text{softmax} \left(\text{mask} \left(\frac{QK^T}{\sqrt{d_k}} \right) \right)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{j'} \exp(e_{ij'})}$$
 formula for softmax

1.00	0.00
0.50	0.50

7. Multiply above Weights with V

(Final Answer) (1 marks)

$$\text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Self-Attention =

5	5
5	5

Question 7. Text Summarization [1+2.5+1+1.5 = 6 Marks]

For the generic text summarization, given below 5 documents in the collection as per the chronological order of mentions, answer the following questions sequentially.

Here are the documents:

Document 1 (d1): "Harry is a brave boy"

Document 2 (d2): "Harry shouted Hermione"

Document 3 (d3): "Hermione shouted Expecto Patronum"

Document 4 (d4): "Ron is a funny boy."

Document 5 (d5): "Harry, Ron and Hermione are friends."

- Preprocess the training document to remove the below set of tokens. Note: Ignore word cases but retain all the words. Stop words & punctuations: { a, are, and, is, '!', ', ' }
- Compute the saliency score of every sentence using the sum of weights (normalized by size of pre-processed sentence) of all the words in the above preprocessed sentences. Use only the new simplified formula to compute word's weightage.

$$\text{Weight (word } W_i) = \text{TF}(i,j) * \text{N/DF}(i)$$

where,

N = Number of document given in the collection

$TF(i,j)$ = Frequency of word i in document j

$DF(i)$ = Number of documents containing word i

- c) Extract summary in reverse chronological order with the top 2 most informative documents from part b)'s result.
- d) Use ROUGE-1 to evaluate the results of the above system summary obtained in part b) w.r.t the below reference summary.

"Harry, Ron and Hermione are good friends"

Solution:

- a) Documents after pre-processing:
 - a. Document 1 (d1): "harry brave boy"
 - b. Document 2 (d2): "harry shouted hermione"
 - c. Document 3 (d3): "hermione shouted expecto patronum"
 - d. Document 4 (d4): "ron funny boy"
 - e. Document 5 (d5): "harry ron hermione friends"
- b) Frequency distribution of words:

		Harry	brave	boy	shouted	hermione	Expecto	Patronum	Ron	funny	friends
d1	harry brave boy	1	1	1	0	0	0	0	0	0	0
d2	harry shouted hermione	1	0	0	1	1	0	0	0	0	0
d3	hermione shouted expecto patronum	0	0	0	1	1	1	1	0	0	0
d4	ron funny boy	0	0	1	0	0	0	0	1	1	0
d5	harry ron hermione friends	1	0	0	0	1	0	0	1	0	1

Weights of words:

		Harry	brave	boy	shouted	hermione	Expecto	Patronum	Ron	funny	friends
d1	harry brave boy	1.667	5	2.5	0	0	0	0	0	0	0
d2	harry shouted hermione	1.667	0	0	2.5	1.667	0	0	0	0	0
d3	hermione shouted expecto patronum	0.000	0	0	2.5	1.667	5	5	0	0	0
d4	ron funny boy	0.000	0	2.5	0	0	0	0	2.5	5	0
d5	harry ron hermione friends	1.667	0	0	0	1.667	0	0	2.5	0	5

Document 1 (d1): "harry brave boy":

Total Weight = $1.667 + 5 + 2.5 = 9.1667$

Sentence Saliency Score (normalized) = $9.1667 / 3 = 3.0556$

Document 2 (d2): "harry shouted hermione":

Total Weight = $1.6667 + 2.5 + 1.6667 = 5.8333$

Sentence Saliency Score (normalized) = $5.8333 / 3 = 1.9444$

Document 3 (d3): "hermione shouted expecto patronum":

Total Weight = $2.5 + 1.6667 + 5 + 5 = 14.1667$

Sentence Saliency Score (normalized) = $14.1667 / 4 = 3.5417$

Document 4 (d4): "ron funny boy"

Total Weight = $2.5 + 2.5 + 5 = 10$

Sentence Saliency Score (normalized) = $10 / 3 = 3.3333$

Document 5 (d5): "harry ron hermione friends"

Total Weight = $1.6667 + 1.6667 + 2.5 + 5 = 10.8333$

Sentence Saliency Score (normalized) = $10.8333 / 4 = 2.7083$

c) Top 2 documents: d3 and d4

Order of summary in reverse chronology = <d4, d3>

d3 = "Hermione shouted Expecto Patronum"

d4 = "Ron is a funny boy."

Summary = "Ron is a funny boy. Hermione shouted Expecto Patronum"

d) Reference summary: "Harry, Ron and Hermione are good friends"

Summary = "Ron is a funny boy. Hermione shouted Expecto Patronum"

Overlapping unigrams = 2

Total unigrams in reference summary = 5

Rouge 1 (recall) = $2/5 = 0.4$

Marking Scheme:

a) 1m: No Partial Marking. Correct Preprocessed document

b) 0.5m: Frequency distribution

1m: Weight of words (for each document)

1m: Sentence Saliency score

c) 0.5m: Correct subset

0.5m: Correct ordering in answer

d) 1m: Rouge calculation

0.5m : Correct value